

Content based Image Authentication with Error Correction

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Abstract: A watermarking scheme for content based image authentication is proposed which, in addition to authentication, has the ability to localize and correct modifications / errors. The watermark is based on the frequency domain features of an image and it is protected using error correcting codes. The use of error correcting codes helps in error localization and correction. The frequency domain features are additionally protected using an approximate message authentication code. This provides tolerance in image authentication when some of the image data cannot be reconstructed due to transmission errors or by legitimate image processing operations such as quantization, compression etc. The minor variations in the image data might not change the content of the image and therefore by correcting some errors, and being tolerant to the errors below a pre defined threshold, results in successful content authentication. Simulation results support that some unintentional data modifications, such as those due to minor channel noise, can be localized and corrected resulting in successful content authentication. Additionally, the simulation results for intentional modifications show the error localization and forgery detection capability of the proposed algorithm.

Keywords - Approximate Authentication; Error Localization; Error Correction; Image Features; DCT; Reed Solomon Codes.

I. INTRODUCTION

Ubiquitous multimedia services have emerged as a result of the rapid growth in Internet and its corresponding communications technologies. Internet helps in easy generation and sharing of images. However, due to the widely available powerful image editing tools, image content can be conveniently edited and perfectly reconstructed, resulting in fake images being accepted as authentic. Consequently, the techniques for image integrity verification and content authentication are becoming ever more significant. Mainly, these techniques are based on digital signatures [1] and / or digital watermarking [2], each having its own pros and cons. Digital signatures based authentication mechanisms require extra bandwidth for the transmission of signatures. In addition, digital signatures are extremely sensitive to modifications. Due to the so called "Avalanche affect" [3], they result in failed authentication in the presence of even a single bit modification, e.g., due to a channel error, or legitimate image processing operations such as quantization, compression etc.

On the contrary, digital watermarking based authentication mechanisms do not require extra bandwidth. They are designed to be prone to minor modifications in the image data. A digital

watermark, calculated on the desired image to be transmitted, is embedded in the cover image and then extracted at the receiver for the verification of integrity and authenticity. Digital watermarking establishes the ownership of an image and helps in verifying its integrity and authenticity.

Traditionally, both of these techniques do not support error localization and correction. Additional mechanisms are needed to locate the exact positions of changes and, if possible, to correct those changes, e.g., due to channel noise, quantization, compression, brightness adjustment etc.

In this paper, a content based image authentication mechanism is proposed, which uses the approximate message authentication code (AMAC) [4] together with error correcting Reed Solomon (RS) codes. AMAC is tolerant to minor changes in data as opposed to the standard message authentication code (MAC) [5, 6] which does not tolerate any modifications in the data. Since AMAC does not provide error correction capability, it is supported in the proposed work using an error correction code, such as RS codes. The frequency domain features of the image are protected using systematic RS codes [7]. RS codes are chosen due to their excellent burst error correction capability. The use of an error correcting code not only helps in the localization of data modifications but also in their correction, if the modifications are within the error correction range of the used error correction codes.

This paper is organized as follows. Section II discusses the related work. Section III introduces the basic concepts used in the proposed scheme. Section IV presents the proposed image authentication mechanism and discusses its error localization and correction capability. Simulation results are given in Section V for intentional and unintentional modifications. Finally, the paper is concluded in Section VI.

II. RELATED WORK

AMAC is an algorithm which provides noise tolerant data authentication. This means if the original data is different from the available data but the differences are below a threshold, the data is accepted as authentic. However, AMAC cannot be directly used for content authentication. Other algorithms for noise tolerant data authentication include approximate image message authentication code (AIMAC) [8], noise tolerant message authentication code (NTMAC) [9], weighted noise tolerant message authentication code (WNTMAC) [10] and error correcting WNTMAC (EC-WNTMAC) [10]. AIMAC is a variation of AMAC, where the AMAC is adapted to image

data in such a way that it is tolerant to minor modifications in the image data but still able to differentiate intentional forgeries. The capability of AIMAC against image modification due to JPEG compression, image forgery and additive Gaussian noise is discussed in [8]. The NTMAC algorithm [9], like AMAC, is tolerant to minor modifications in data. In NTMAC, the idea is to split the data into blocks, calculate a standard MAC on each block but retain only a portion of the whole MAC for that block. This portion is used for detecting modifications in that block. Partitions are introduced for tolerance against minor modifications. WNTMAC [10] is based on the NTMAC algorithm and introduces the concept of weights for differentiating the important parts of data from the less important parts. NTMAC and WNTMAC do not provide error correction capability. EC-WNTMAC [10] is an extension of WNTMAC, where the error localization and correction capability is introduced aside from error tolerance. All of the above mentioned approaches are used for authentication of image data rather than its content.

Error correcting codes have been used in literature, in combination with watermarking, for error location / localization and correction. In [11], RS codes are used to generate parity symbols for each row and column of the data of an image, which are embedded as a watermark in the two least significant bit planes of the image. RS decoder is used to correct the errors / modifications in the watermarked image and to reconstruct the data of the original image. A watermarking approach is proposed in [12], in which an image is divided into blocks and the block hash is encoded with a systematic error correcting code. The parity symbols are then embedded into the blocks. During verification, the hash of each block is recovered from the embedded parity, if the number of tampered blocks stays below a predefined threshold.

Two techniques for image content protection using self-embedding an image in itself are introduced in [13]. This helps in recovering portions of images which are cropped, replaced, damaged or tampered. In the first method, the 8x8 blocks of an image are transformed into frequency domain using discrete cosine transform (DCT) and the coefficients are embedded in the least significant bits of another block. This method is fragile but it has a good quality of reconstruction. The second method is based on the principle of differential encoding where a circular shift of the original image with decreased color depth is embedded in original image.

To the best of our knowledge, there is no watermarking scheme in literature which performs noise tolerant image content authentication and at the same time uses error correcting codes for the detection of modifications, localization of the region of modifications and correction of the modified features. One such scheme is proposed in [14] by the authors of this work but it is not based on digital watermarking. The scheme of [14] is called two phase image authentication scheme, where data is protected in two phases by generating two authentication tags on the image. The low frequency image features are processed in the first phase and if the first phase succeeds, then the high frequency elements are processed in the second phase. If the first phase fails, the image content is considered unauthentic. If both phases succeed together, the image content is considered authentic.

III. BUILDING BLOCKS OF THE DUAL WATERMARKING SCHEME

A. Discrete Cosine Transform (DCT)

DCT of an image makes a spectral analysis of the image and orders the spectral regions from high to low energy. DCT is either applied to the whole image or by splitting the image into smaller blocks. DCT of an N×N block is calculated as given by (1),

$$F(u, v) = c(u)c(v) \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} f(i, j) \cos\left[\frac{\pi(2i+1)}{2N}u\right] \cos\left[\frac{\pi(2j+1)}{2N}v\right] \quad (1)$$

$$0 \leq u \leq N-1, 0 \leq v \leq N-1$$

$$c(u) = \begin{cases} \sqrt{\frac{1}{N}}, u = 0 \\ \sqrt{\frac{2}{N}}, u \neq 0 \end{cases}, c(v) = \begin{cases} \sqrt{\frac{1}{N}}, v = 0 \\ \sqrt{\frac{2}{N}}, v \neq 0 \end{cases}$$

The inverse DCT transform of each block is obtained from the frequency components using (2),

$$f(i, j) = \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} c(u)c(v) F(u, v) \cos\left[\frac{\pi(2i+1)}{2N}u\right] \cos\left[\frac{\pi(2j+1)}{2N}v\right] \quad (2)$$

$$c(u) = \begin{cases} \sqrt{\frac{1}{N}}, u = 0 \\ \sqrt{\frac{2}{N}}, u \neq 0 \end{cases}, c(v) = \begin{cases} \sqrt{\frac{1}{N}}, v = 0 \\ \sqrt{\frac{2}{N}}, v \neq 0 \end{cases}$$

B. Reed Solomon Codes

Reed Solomon (RS) codes are cyclic block codes defined over Galois Field (GF). A systematic RS encoder encodes a message of k symbols defined over GF(q = 2^m), where 2 ≤ m ≤ 8. It appends the generated (n - k) parity symbols to the data symbols to produce a codeword of n symbols. RS decoder can detect and correct symbol errors, such that it can detect and locate (n - k) symbol errors.

RS codes can also either correct (n - k) / 2 symbol errors or (n - k) symbol erasures. Erasures are those errors whose positions are known in advance, e.g., via channel reliability information. RS codes are also used to simultaneously correct a combination of symbol errors and erasures. The symbol erasure correction capability of RS codes is used in the approach presented in this paper.

C. Approximate Message Authentication Code (AMAC)

AMAC is a special cryptographic construct which is designed to authenticate messages but tolerate minor modifications in them, unlike the standard MAC which does not tolerate even a single bit modification. The threshold on the acceptable number of errors is adaptable and the tolerance in AMAC is achieved based on the majority logic.

AMAC tag generation [4] on a message M is shown in Fig. 1, where the length of the generated tag is L. Here, R and S are positive integers, where R is usually chosen to be equal to S for simplicity [4]. A pseudorandom bit stream generator is used in the AMAC which is based on a shared key, known to both the sender and the receiver, so that the same pseudo random sequence can be generated on both sides.

The majority logic used in the AMAC algorithm helps in achieving a level of tolerance against modifications. As long as the majority of a particular row (or column) is not changed, the corresponding bit in the AMAC tag remains unchanged, resulting in a certain degree of tolerance.

IV. PROPOSED APPROXIMATE AUTHENTICATION SCHEME WITH ERROR LOCALIZATION AND CORRECTION

A. Transmitter Side

The input image (I) is divided into non overlapping blocks of 8×8 pixels. The DCT of each block is computed to produce as many DCT matrices as the number of blocks obtained from the image. The DC elements of the DCT, of each block, are chosen for protection with RS codes and watermark generation as they contain the most useful information. However, it is to be noted that any other features of the image are also valuable and can be used instead of the DCT. Each DC element is 16-bit long.

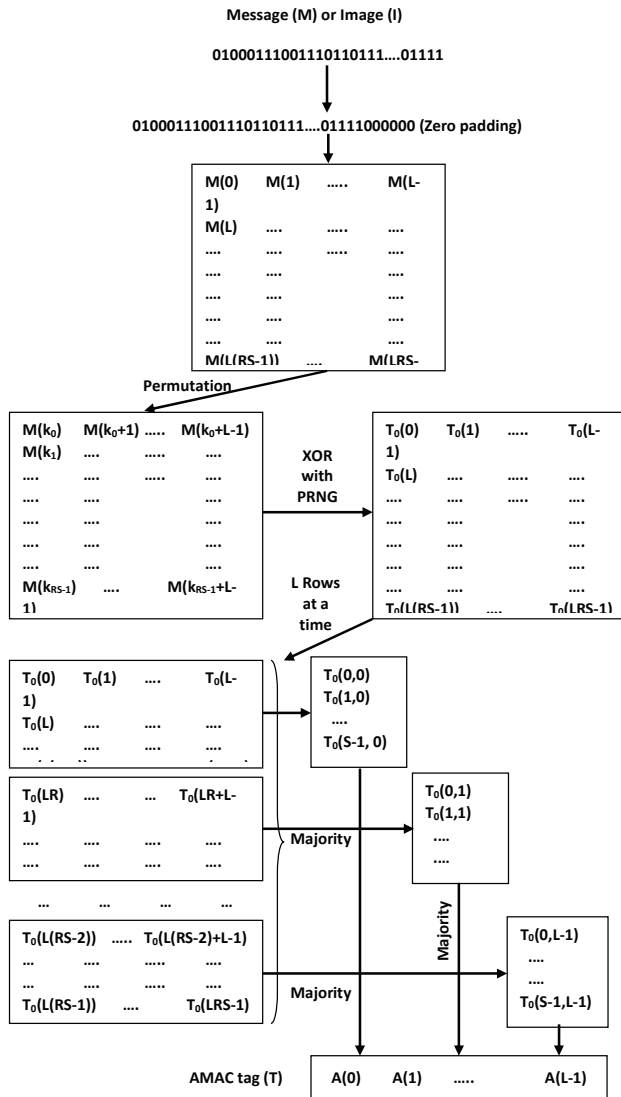


Figure 1. AMAC tag generation [4]

In the proposed scheme, the DC elements are protected using RS(255, 239) to provide 8 symbol error correction capability. However, any other code length or in general any other error correction code can be used. The RS(255, 239) code is shortened according to the image resolution for flexibility and adaptability of the code to the image resolution. For example, for a 128×128 pixel image with 8×8 blocks, there are 16 blocks in each row and therefore 16 DC elements are to be protected with RS(255, 239) in each row. However, since the number of DC elements is too low for RS(255, 239), an RS(48, 32) code, shortened from RS(255, 239) is chosen to provide an error correction capability of 8 symbol-errors. Each 16-bit DC element is split into two 8-bit RS symbols and therefore up to 4 erroneous DC elements can be corrected by the 8 symbol error correction capability of the code. However, by using an erasure correcting RS decoder, this capability is extended to 8 DC elements. That is the reason for using erasure correcting RS codes in the work proposed in this paper.

The DC elements of all the blocks are arranged in rows and RS encoding is done one by one over each row to obtain the RS parity symbols as shown in Fig. 2. 16 RS parity symbols (RS-Parity_i) are obtained for each row, i , of blocks where $1 \leq i \leq N$ and N is the number of blocks in a column. This approach provides protection to every DC element and helps in localizing the errors to the block level. It is also able to correct the identified / marked erroneous DC elements. After all the parity symbols have been calculated, the RS-Parity for each row is embedded into a different row. The destination row is chosen randomly using a pseudo-random number generator function.

The pseudo random numbers are obtained using a key as a seed value, which is chosen to be the same key as used for AMAC tag generation in the proposed work, however, in general it should be a different shared key. The key helps in obtaining the same pseudo random sequence for inverse permutation at the receiver side. This is done to avoid the destruction of the features and their protection (parity) symbols at the same time if a row of blocks in the image is affected by noise or intentional modifications. The parity symbols are converted into their equivalent bits values. These bits are then equally distributed over all the blocks depending on the image resolution and embedded into the least significant bit of the image pixel values in the block $B_{i,j}$ in sequence, starting from the first pixel till as long as there are bits to be embedded. The permuted storage of the watermark is depicted in Fig. 3.

dc _{1,1}	dc _{1,2}	dc _{1,3}		dc _{1,N}	RS-Parity _{r1}
dc _{2,1}	dc _{2,2}	dc _{2,3}		dc _{2,N}	RS-Parity _{r2}
dc _{3,1}	dc _{3,2}	dc _{3,3}		dc _{3,N}	RS-Parity _{r3}
.....					
.....					
dc _{N,1}	dc _{N,2}	dc _{N,3}		dc _{N,N}	RS-Parity _{rN}

Figure 2. DC protection using RS codes

The DC elements of all the blocks are combined and passed to the AMAC tag generation algorithm to generate the AMAC tag T . This tag is transmitted together with the image and used on the receiving side for (approximate) content authentication of the image. The AMAC used in this work has a slight modification over the original AMAC as proposed in [4]. It uses an extra pseudo random bit for AMAC computation in order to avoid the security attack proposed in [15]. Whereas the original AMAC uses one pseudo random bit XORed with the message bit, the modified method uses two pseudo random bits, using AND ($\cdot$) and XOR ($\oplus$) operations with the message bit as given in (3).

$$O_i = M_i \cdot R1_i \oplus R2_i \quad (3)$$

where M_i is the i^{th} message bit, $R1_i$ and $R2_i$ are the corresponding pseudo random bits and O_i is the output bit.

B. Receiver Side

The image (I') is received together with its AMAC tag (T'). Either the image or its tag (or both of them) might be effected by unintentional modifications, such as channel noise or image processing operations, or by intentional modifications, such as forgery attacks. The received image (I') is split into 8×8 pixel non overlapping blocks. The watermark bits are extracted first to be used as parity bits. An inverse permutation, of the permutation at the transmitter side, is used to extract the watermark bits. The key for inverse permutation is the same as the key used for permutation. These bits are combined together to make the parity symbols over GF, which are used in the decoding of RS codes. The DCT of each block is calculated and the DC elements are combined together with the parity symbols for that row of blocks which are extracted using the inverse permutation. RS decoding is performed on each row of the DC elements.

The RS decoding results in three possibilities, i.e., decoding succeeds, there is a decoding error or there is a decoding failure. Decoding succeeds when the decoder corrects the erroneous symbols / erasures. Decoding failure occurs when the decoder is not able to correct the errors / erasures, e.g., when there are more errors than the error correction capability of the code. Decoding error occurs when the decoder thinks it has corrected the codeword, but in fact it has wrongly corrected the erroneous symbols in the received word without realizing it. Decoding error is also widely known as miscorrection.

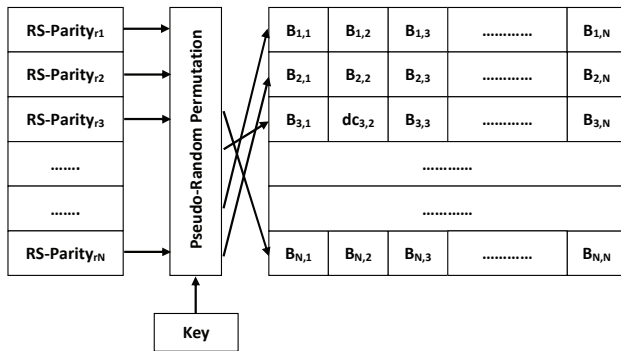


Figure 3. DC protection using RS codes

Miscorrections are possible when the received word is at the same Hamming distance from two valid codewords and the decoder makes a wrong choice. It is not possible to differentiate between a decoding error and a successful decoding without using an external error detection code, such as a CRC code.

In any case, the error locator polynomial for each RS codeword, i.e., each row of blocks will identify the potentially erroneous DC elements and therefore the corresponding potentially erroneous image blocks. Some of these DC elements are correctable, if the total number of errors in the row of DC elements does not exceed the error correction capability of the chosen RS code. If the DC element of a block is identified to be in error and not correctable, then the corresponding block is marked as potentially wrong. Otherwise the block is taken as correct or authentic. However, for the whole image to be considered as authentic, all the DC elements must also pass the additional AMAC tag verification step as discusses next.

After the RS decoding of each row of DC elements, the DC elements are combined and passed through the modified AMAC algorithm used in this work. This produces the AMAC tag T'' . This tag is compared with the received tag T' . If the two tags are equal, the AMAC algorithm authenticates that the DC elements are either similar or have minor differences, which might be tolerable. The tolerance can be chosen as desired by adjusting the R parameter used in AMAC. This step not only authenticates the content but also verifies that the results of the RS decoding are correct. If the RS decoder has resulted in a decoding success but the AMAC tag verification fails, it implies there must have been decoding errors and the image is not authentic.

V. SIMULATION RESULTS

The simulation parameters are chosen as follows. The images chosen for simulations are grayscale with a resolution of 128×128 pixels. The images are divided into blocks of 8×8 pixels, thus there are 16 rows (and columns) of blocks in each image. The length of AMAC tag is chosen as 256-bits. Systematic RS(255, 239) codes are used, which are shortened to RS(48, 32) due to the image resolution as explained earlier. Each 16-bit DC element is represented by two RS symbols, since the RS symbols are 8-bits each due to the used GF(8).

Fig. 4(a) shows an original Lena image of the chosen resolution. Fig. 4(b) shows the Lena image in the presence of "Salt & Pepper" noise of magnitude 0.01. Fig 4(d) shows that the RS decoder is able to identify the (potentially) erroneous blocks, which are marked as red. As the noise level is low, the RS decoder is able to correct most of the DC elements in the rows of blocks. The authentication is successful because the proposed authentication scheme is able to correct the modifications in DC elements and also tolerate any remaining modifications by virtue of the used AMAC authentication algorithm. Fig. 4(c) shows the modified Lena image in the presence of "Salt & Pepper" noise of magnitude 0.1.

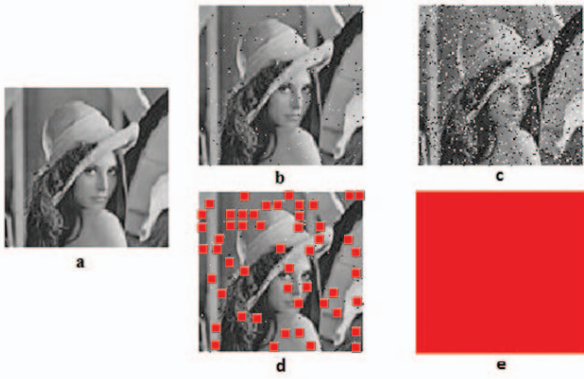


Figure 4. Authentication in the presence of "Salt & Pepper" noise

Since the noise level of Fig. 4(c) is much higher as compared to the noise level of Fig. 4(b), the identified modifications are much higher as shown in Fig 4(e), which is all red. This means that all the blocks are marked as potentially erroneous. However, the large amount of modifications is beyond the error correction capability of the chosen RS code. The modifications are also not tolerable by the AMAC authentication algorithm. Thus the received image is declared unauthentic by the proposed content based image authentication scheme.

Fig. 5(a) shows a forged Lena image, i.e., it is a case of intentional modifications. The forged area can be seen in the forehead part where extra hair is visible. The forged / modified blocks are identified using the error locator polynomials of the RS decoder and are shown in red in Fig. 5(b). Since the forgery / modification is concentrated in one region and the amount of modification is too high, the RS decoder fails to correct it despite using its erasure correction capability. At the same time, due to the usage of majority logic in AMAC, the tag T'' recalculated by AMAC comes out to be different than the received tag T' . Thus the authentication fails and the image is declared as unauthentic.

This example shows that intentional modifications / forgeries might or might not be correctable, but it might still be possible to identify the forged locations in the image.



Figure 5. Authentication in the presence of forgery attack

VI. CONCLUSION

A novel content based image authentication scheme is presented in the paper, which provides error localization and also correction to a certain extent as supported by the used error correcting code. The scheme consists of generating RS parities over the DC elements of image blocks in each row. The RS parity symbols of each row are embedded as watermark into different rows of the original image using a pseudo random permutation. An AMAC tag is generated over the DC elements. If the RS decoding and AMAC tag verification succeeds together, the content of the image is considered authentic. Otherwise the image is declared un-authentic. Simulation results show that unintentional modification below a certain threshold are identified and corrected by the proposed approach. Intentional forgeries are not correctable but it might still be possible to identify the forged area in the image.

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